

DBSCAN

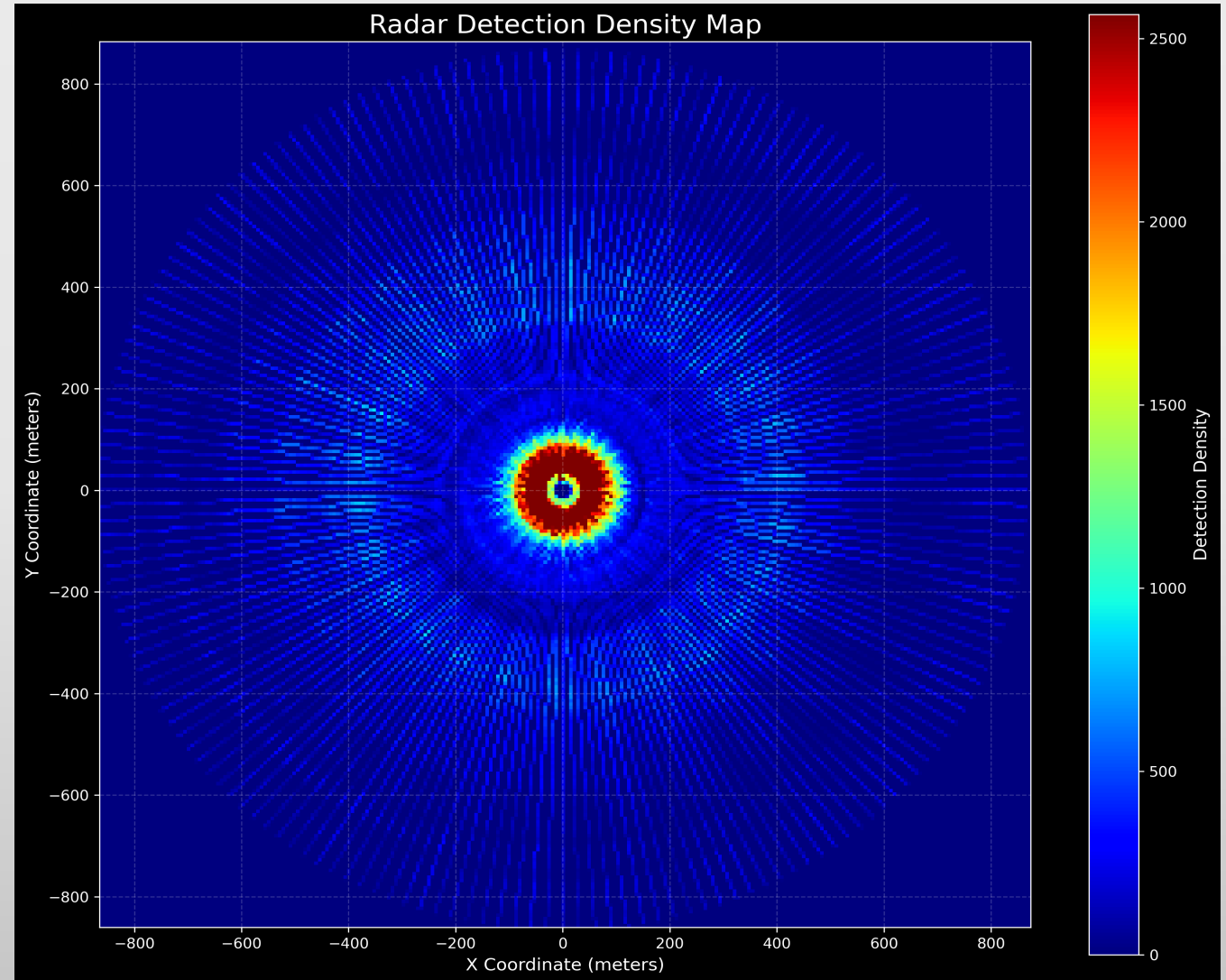
Sam Cancilla 9-16-2025



Where We Left Off

Challenges we had

- Plotted data was obscure
- Processing the data was sluggish
- When fixed the image with DBSCAN was not fully shown
- Plotted radar data was sparse and not detailed



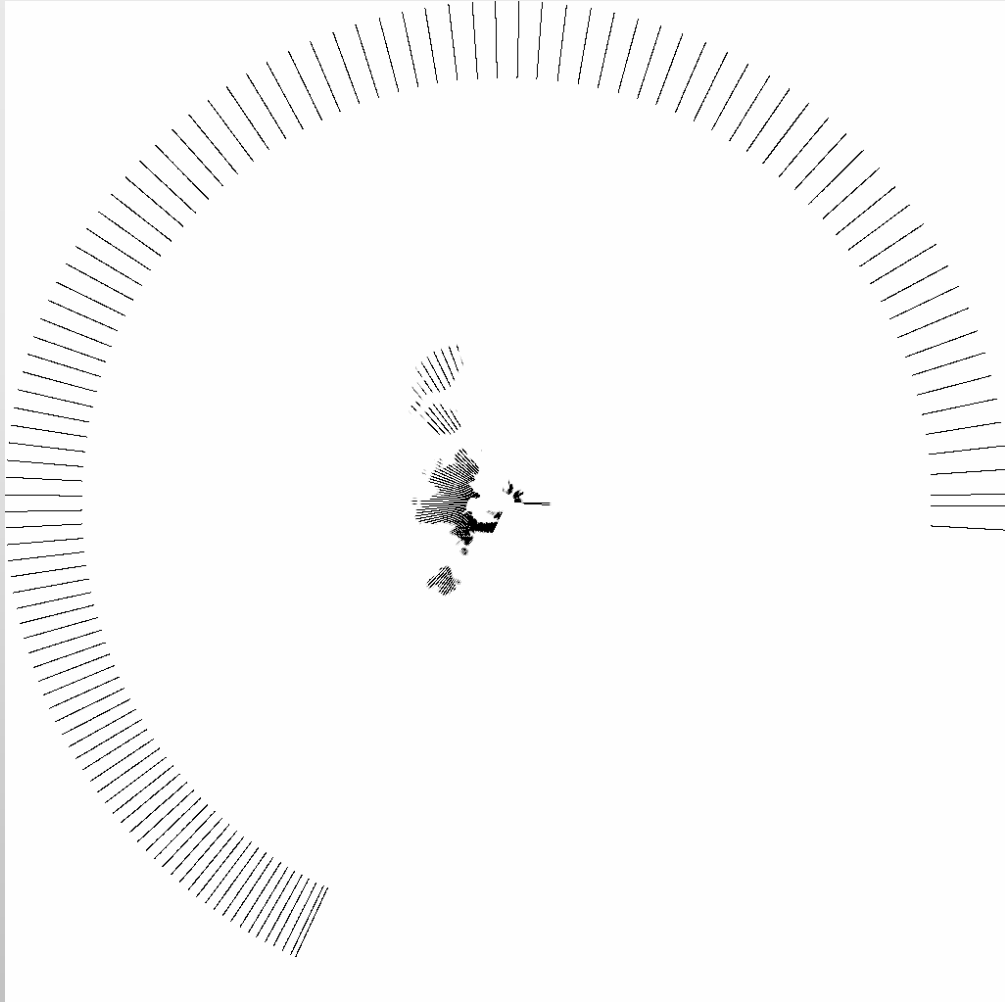
Fixed DBSCAN

CSV to PNG

- The CSV to PNG is now much more detailed
 - Coves are more visible
- Able to see how reflective things are via. Brightness
- Based the new DBSCAN off of the reading CSV to PNG



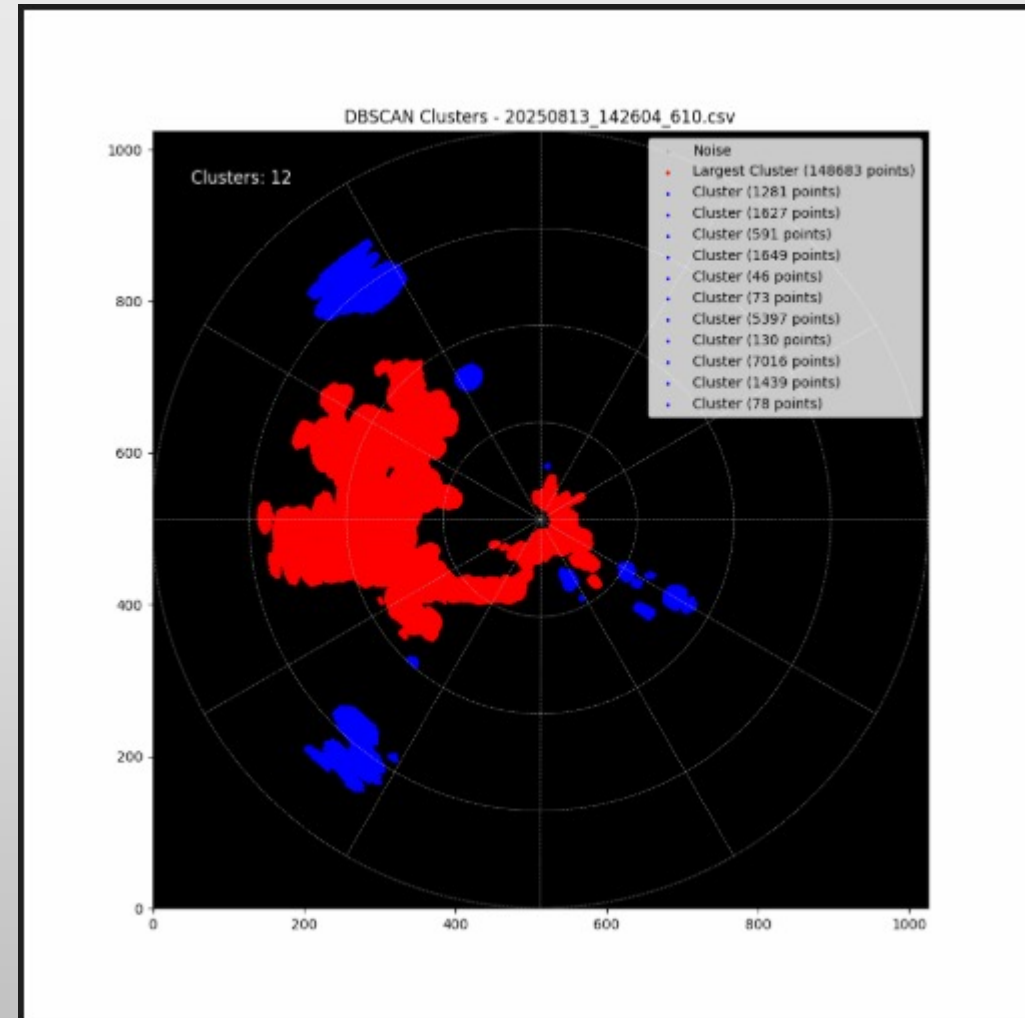
Side by Side Comparison



DBSCAN Processed Images

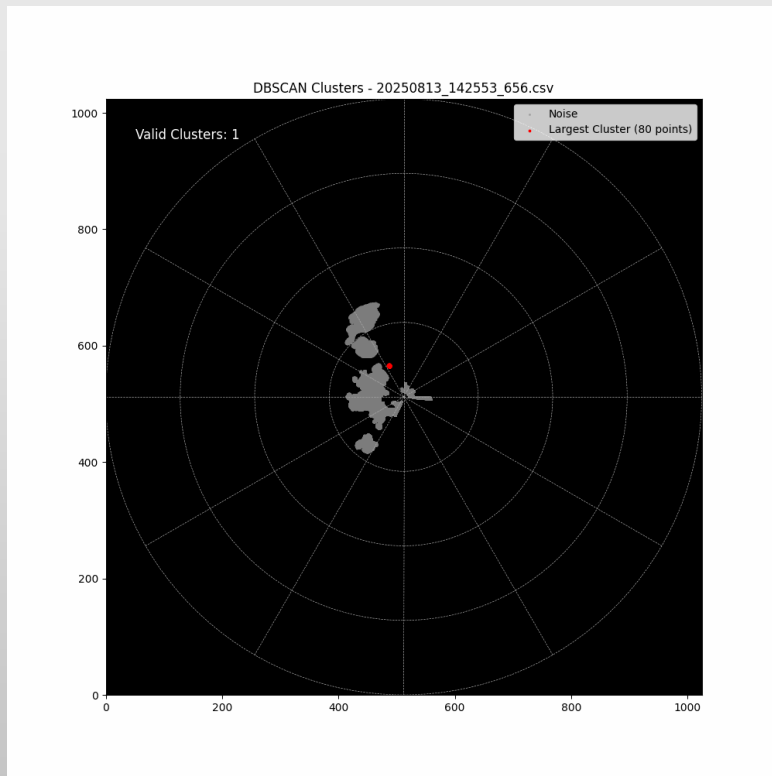
- When generating there are variables that we can change
 - Epsilon
 - Min Samples
 - Min Cluster Size
 - Max Frames
 - Output File Name
- With current parameters land is detected as noise
 - Can have it so land is detected as a very large cluster

```
PS C:\Users\saman\Desktop\SamPython> Python C:\Users\saman\Desktop\SamPython\DetectTargetsNONML\@dbscan_animation_generator.py --eps 12 --min_samples 4 --min_cluster_size 15 --max_frames 10 --output_filename "Whatever_Name.gif"
```

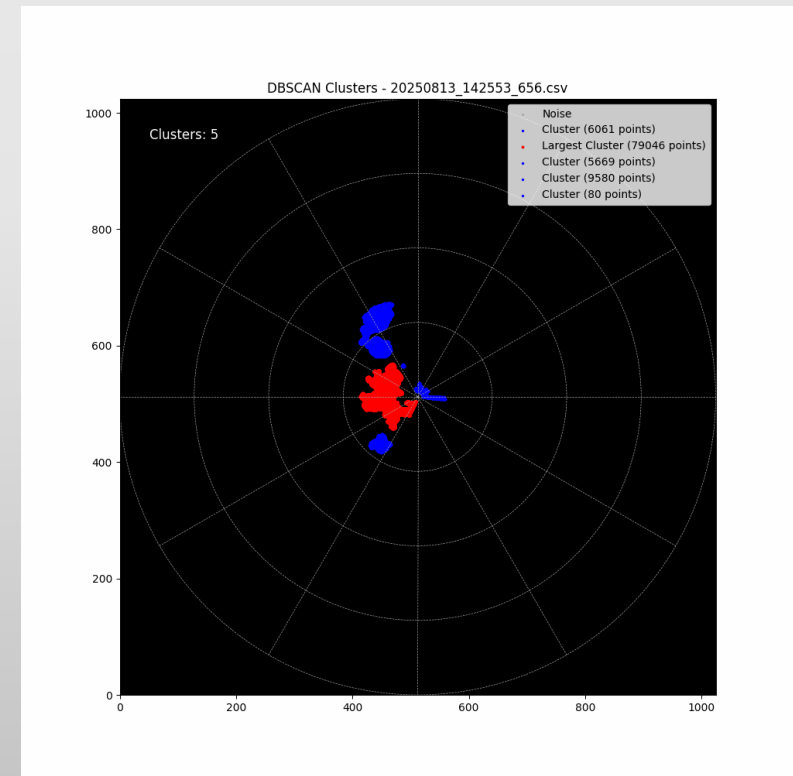


Side by Side

More restrictive DBSCAN

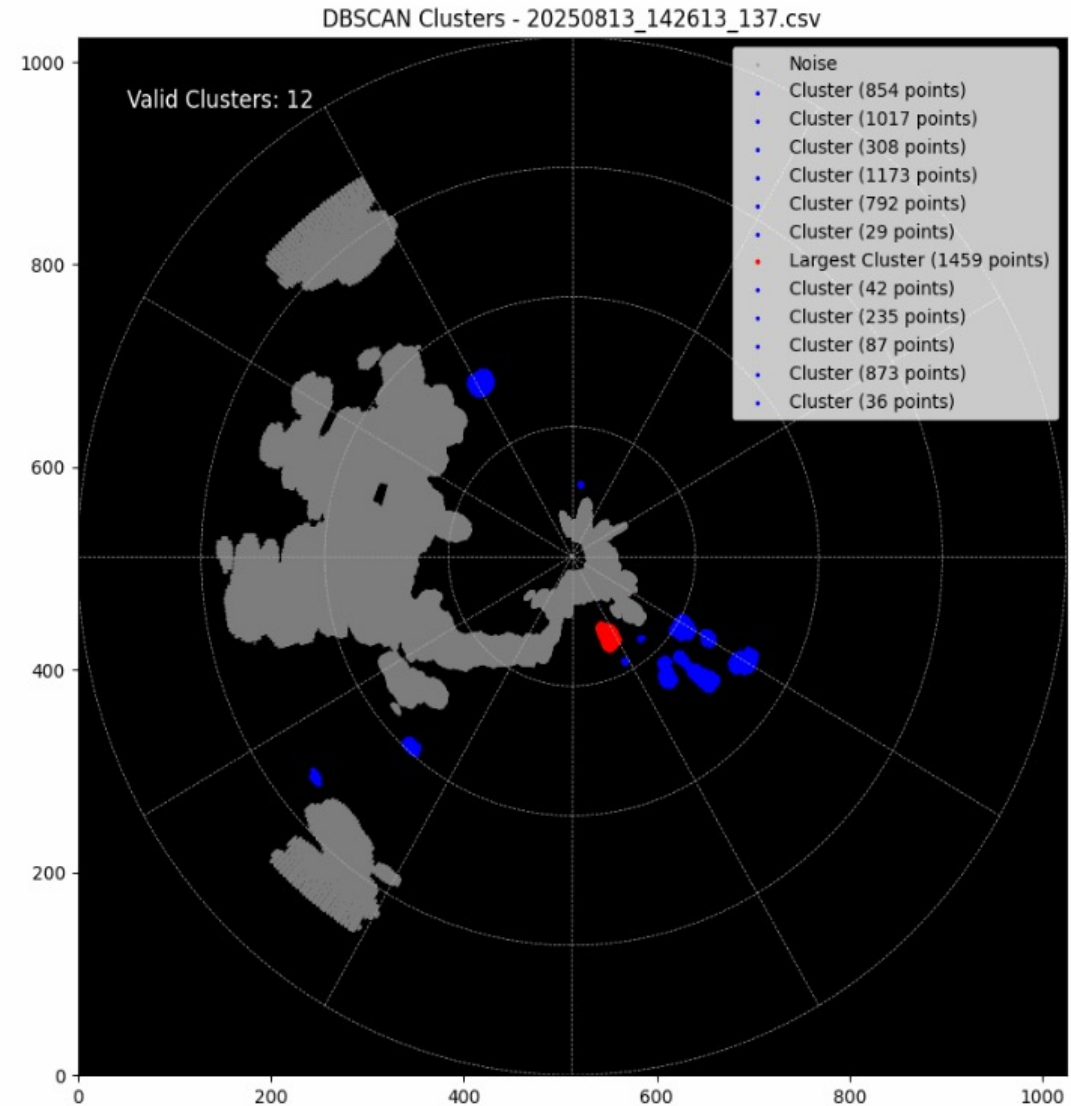


Less restrictive DBSCAN



Implementing Tracking

- I think the easiest would be using the filtered DBSCAN
 - Every cluster is essentially labeled as an object of interest
 - Haven't tried it with other datasets
- Refine and try more filters on different datasets to find best fit



Moving forward

- Get tracking up and running
 - Jacob – What method of DBSCAN would be best for tracking?
- DBSCAN across multiple frames to fix disappearance
- Start Point Cloud



The background of the slide is a vibrant blue, densely populated with numerous speech bubbles of various colors including red, yellow, pink, and white. Each speech bubble contains a large, bold, dark blue question mark. The bubbles are scattered across the entire frame, creating a pattern that suggests a continuous stream of questions or inquiries.

Thank you.

Any Questions?